

Study on the impact of board gender diversity and skill ratio on firm's market capitalization and short-term debt in the pre-2008 financial crisis scenario

Introduction

The purpose of this study is to determine the impact of board composition and workforce skills on a firm's performance. The specific research question is: To what extent does internal diversity in board composition and workforce skills contribute to variation in firms' market capitalization and exposure to short-term risky debt? The motivation stems from an interest in certain corporate governance and organizational behavior theories. The Agency Theory (Jensen and Meckling, 1976) argues that boards are responsible for monitoring management and protecting shareholder interests. The increase of gender diversity can strengthen this oversight by broadening perspectives and reducing managerial opportunism. Empirical evidence supports this mechanism: Adams and Ferreira (2009) prove that gender-diverse boards improve oversight and performance, particularly in firms with weaker governance structures. Recent findings further show that female directors are associated with improved financial performance across both favourable and adverse market conditions (Basuony et al., 2023) and tend to adopt more risk-averse financial strategies, including reduced reliance on debt financing (Sharma, 2024). According to Social Identity Theory (Tajfel, 1978), diverse groups are less likely to groupthink and expected to engage in more careful and balanced decision-making, which can translate into more prudent financing strategies. Human Capital Theory (Becker, 1964) suggests that highly skilled employees increase organizational productivity and innovation capability, which should improve long-term financial performance and firm valuation. Empirical evidence is consistent with this view: firms with a higher proportion of skilled employees tend to exhibit stronger productivity and higher investment efficiency by reducing both overinvestment and underinvestment (Chen, 2023). This suggests that skill intensity contributes not only to value creation but also to greater financial stability. Together, these theories imply that firms with more diverse boards and more skilled employees should experience higher market capitalization and lower dependence on risky short-term debt.

To address the research question, panel data estimation methods are employed, namely fixed-effects and random-effects estimators. This choice is motivated by the panel structure of the dataset, which follows approximately 4.000 firms during the two years immediately preceding the 2008 global financial crisis, a particularly sensitive period for corporate financial decisions, as well as from the well-known limitations of the pooled OLS estimator. In particular, pooled OLS fails to account for firm-specific heterogeneity: if time-invariant firm characteristics are correlated with the regressors, pooled OLS yields biased and inconsistent estimates. Moreover, the estimator does not control for common time shocks such as macroeconomic conditions that might affect regressors and outcomes simultaneously, thereby amplifying the risk of omitted variable bias. In addition, because it ignores that the same firm is observed repeatedly, pooled OLS cannot accommodate serial correlation in the errors, leading to underestimated standard errors. It also mixes within- and between- firm variation, thus preventing meaningful interpretation of dynamic relationships. Finally, the OLS estimator cannot address potential reverse causality, such as the possibility that firms with stronger financial performance attract more diverse boards and skilled employees, thereby introducing further endogeneity concerns.

Table 2 reports the pooled OLS estimates for log market capitalization and centered short-term debt under both simple and extended specifications. For log market capitalization, board gender diversity does not show a statistically significant effect in any specification. In contrast, the skill ratio is negatively associated with firm financial size in both simple and extended models: a 10-percentage-point increase in the share of skilled workers is associated, on average, with a 4.6% decrease in market capitalization,

even when sector and firm longevity are controlled for. For centered short-term debt, results differ markedly. Gender board diversity is strongly and consistently associated with lower short-term debt levels. The coefficient is large and statistically significant in both the simple and extended regression: a 10-percentage-point increase in the share of women on boards is associated, on average, with a 0.24 million decrease in centered short-term debt. Keeping sector and firm age constant, a smaller but still meaningful decrease of about 0.15 million on average is detected. Skill ratio, on the contrary, does not exhibit any significant relationship with short-term risky debt.

For the reasons outlined above, the pooled OLS estimates should not be interpreted as causal effects. Panel data methods will be used to control for unobserved firm-specific, time-invariant characteristics as well as common macroeconomic shocks and exploit both the cross-sectional and time-series dimension of the data, thereby improving identification of the causal impact of internal board diversity and workforce skill composition on financial outcomes.

Data structure and methodology

Prior to estimation, an exploratory analysis was carried out to examine the main financial and organizational indicators in the dataset. Table 3 reports summary statistics for variables such as market capitalization, short-term debt, investment, profits, board gender composition and workforce skill levels. The results reveal substantial heterogeneity in both outcome variables. Market capitalization ranges from 1.23 to 269.45 million, confirming the presence of both small and large firms in the sample. In contrast, short-term debt displays a tighter range (15.82–26.74 million), indicating lower dispersion across firms yet sufficient within-sample variation to justify econometric investigation. To improve interpretability, the investigated outcomes are expressed as the logarithm of market capitalization, making the distribution more comparable across firms and reducing the influence of extreme values, and as centered short-term debt, which captures deviations from the average firm level and reduces potential multicollinearity. The distributions of both variables were graphically examined using histograms. The log-transformed market capitalization histogram shows a unimodal, left-skewed distribution with the highest bars clustering on the right side of the graph. This shape indicates that while the modal firm size is relatively large, many smaller firms with progressively decreasing market capitalization values remain present in the sample (Figure 1). Similarly, the histogram of average centered short-term debt displays a unimodal, roughly symmetrical distribution, with the highest bars clustering around the center (Figure 2). This pattern is consistent with the mean-centering transformation, since most firms accumulate levels of short-term debt close to their long-run average.

Regarding the explanatory variables of interest, both gender diversity and skill ratio exhibit meaningful variation across firms. Gender diversity ranges from 0 to 94%, with a mean of 12% and a median of 6.7% (Table 3). The mean value exceeding the median is consistent with the right-skewed distribution displayed in the histogram (Figure 3). This reveals that while the sample boards tend to be male-dominated, sufficient heterogeneity across firms is present to study the effect of diversity. The share of skilled employees, on the other hand, is fairly uniform across the central region of the support. Values go from 0 to 17% (Table 3), with a mean value of around 7.6%. Consistent with the kernel density plot, the mass of the distribution lies between 4% and 11%, with thinner tails (Figure 4). This indicates substantial cross-firm variation in workforce skill intensity, with no evidence of concentration around specific levels. Having established the main distributional features of the governance and workforce measures, the analysis now turns to estimating their impact on firms' performance and financial decisions. To obtain valid inference in the presence of potential within-firm serial correlation and heteroskedasticity, all models are estimated with cluster-robust standard errors at the firm level. Accordingly, two firm fixed-effects models

are implemented with the following specifications:

$$\begin{aligned}\ln(mktcap_{it}) &= \beta_1 genderdiversity_{it} + \beta_2 skillratio_{it} + \beta_3 age_{it} + \mu_i + u_{it} \\ shortdebtcentrage_{it} &= \gamma_1 genderdiversity_{it} + \gamma_2 skillratio_{it} + \gamma_3 age_{it} + \mu_i + \varepsilon_{it}\end{aligned}$$

Identification is achieved from within-firm changes over time, as fixed effects difference out all the time-invariant firm characteristics. Firm longevity is included as a control in the specification, as it varies over time and is compatible with the within-firm fixed effects estimator. Other potential control variables, such as profit and investment, were deliberately excluded from the analysis, as they are endogenous outcomes of firms' strategic decisions and not predetermined with respect to the dependent variable. Including them would introduce potential bias through reverse causality. Similarly, sector was not retained as a control in this first stage because it is time-invariant and therefore fully absorbed by the firm fixed effects.

To assess the robustness of the result, the specification is extended to include time fixed effects. This controls for aggregate shocks, business cycle dynamics and common macroeconomic conditions, including those associated with the lead-up to the global financial crisis, which affected all firms simultaneously:

$$\begin{aligned}\ln(mktcap_{it}) &= \beta_1 genderdiversity_{it} + \beta_2 skillratio_{it} + \mu_i + \lambda_t + u_{it} \\ shortdebtcentrage_{it} &= \gamma_1 genderdiversity_{it} + \gamma_2 skillratio_{it} + \mu_i + \lambda_t + \varepsilon_{it}\end{aligned}$$

Firm age is excluded from the specification because it is perfectly collinear with the year fixed effects: the mechanical annual increase in age is entirely absorbed by the time dummies. Year fixed effects are included for 2007, with 2006 serving as the reference category. With both firm and time fixed effects included, identification comes from idiosyncratic within-firm changes over time, net of any year-specific macroeconomic trend. In practice, the estimates reflect deviations from each firm's own average and from the aggregate shocks common to all firms in a given year.

Figure 5 and 6 show that in the sample there is almost no change in gender board composition going from 2006 to 2007 and changes in skill ratio are of very small magnitude. This indicates that the primary source of variation in these variables is cross-sectional rather than within-firm, which limits the identifying power of the fixed-effects specification. For this reason, both within- and between- firm variation is exploited by extending the model to a random effects specification, which incorporates sectoral differences while retaining cross-sectional variation. In fact, figure 7 and figure 8 reveal differences in level and dispersion across sectors, indicating that sector affiliation contributes to cross-sectional heterogeneity that would be absorbed under the fixed-effects estimator.

$$\begin{aligned}\ln(mktcap_{it}) &= \beta_1 genderdiversity_{it} + \beta_2 skillratio_{it} + \beta_3 age_{it} + \sum_{s=2}^{10} \delta_s D_{is} + c_i + u_{it} \\ shortdebtcentrage_{it} &= \gamma_1 genderdiversity_{it} + \gamma_2 skillratio_{it} + \gamma_3 age_{it} + \sum_{s=2}^{10} \theta_s D_{is} + c_i + \varepsilon_{it}\end{aligned}$$

Exploiting both sources of variation, identification here relies on cross-sectional differences across firms as well as on changes over time. The random effect c_i is assumed to be uncorrelated with the regressors, crucial for the consistency of the random-effects estimator. Sector dummies D_{is} are included to capture systemic differences in market capitalization and short-term debt across industries, with sector 1 as the omitted baseline. To assess the plausibility of the random-effects assumptions and ensure that unobserved heterogeneity is uncorrelated with the regressors, a Hausman test was conducted. To determine the estimates' consistency, the test formally compares the fixed-effects and random-effects estimators for both outcome variables under the null hypothesis that the coefficients are equal. If the resulting p-value is not significant, the null hypothesis cannot be rejected, and the random-effects specification is preferred

as it exploits more variation, thereby yielding more efficient estimates. Conversely, a significant p-value indicates inconsistency in the random-effects specification, likely due to correlation between the regressors and the firm-specific effect, thus favoring the fixed-effects model.

Model results

Table 4 displays the results of the fixed effects model with a one-way specification. After controlling for firm-specific fixed characteristics, changes in gender diversity, skills, and firm longevity do not explain changes in market capitalization, as none of the estimated coefficients is significant. Similarly, centered short-term debt is not significantly affected by gender diversity or skill ratio. An interesting and significant result emerges for firm longevity. Within a given firm, an additional year of experience is associated with an increase of about 1.02 million in centered short-term debt, on average, after accounting for all time-invariant firm characteristics. Although the within- R^2 suggests that the fixed-effects model captures substantial within-firm variation over time, this explanatory power does not come from the social-composition variables of interest. When time fixed effects are added to the model specification, these results are qualitatively identical, confirming model robustness. Once time fixed effects absorb the firm's longevity pattern, the positive 2007 coefficient implies that, on average, firms' centered short-term debt was uniformly higher in 2007 than in 2006. Thus, the earlier age effect appears to have captured a common macroeconomic shift rather than true within-firm aging effect.

Table 1: Two-way firm and time fixed effects regression results

<i>Dependent variable:</i>		
	Log market cap (1)	Short-term debt (2)
Gender diversity	0.009 (0.016)	-0.869 (2.543)
Skill ratio	0.019 (0.020)	1.627 (3.689)
Year 2007	0.0002 (0.0001)	1.021*** (0.019)
Observations	8,000	8,000
R ²	0.001	0.428
Adj. R ²	-1.000	-0.144

Notes: Firm and year fixed effects included; age omitted due to collinearity. Standard errors clustered at the firm level. Centered short-term debt is mean-centered and expressed in millions. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5 confirms that no impact is detected for gender diversity and skill ratio on market capitalization, even when between-firm variation is incorporated through the random-effects specification. Even though most sector coefficients are statistically insignificant, sector seven presents an interesting result. Conditional on the model's specifications, firms in this sector have, on average, lower market capitalization relative to those in sector one. This pattern indicates that sector seven is potentially less capital-intensive or operates with smaller firm scales. More statistically significant results arise for short-term debt. Gender diversity appears to have a negative association: a 10-percentage-point increase in the share of women on the board is associated, on average, with a reduction in centered short-term debt by approximately 0.155 million, conditional on skill intensity, firm age and sectors. Several sector coefficients are also significant, suggesting that short-term financing practices differ systematically across industries. However, these patterns cannot be interpreted as reliable causal within-firm effects. The Hausman test rejects the random-effects estimator for centered short-term debt, implying that unobserved firm characteristics

are correlated with the regressors, potentially biasing the analysis. As a consequence, the fixed-effects model remains the only valid estimator for this outcome. By contrast, the random-effects model is not rejected for log market capitalization, indicating that the estimator is consistent and therefore the sectorial difference found is reliable (Table 6).

Findings, heterogeneous effects, robustness checks and analysis limitations.

No robust evidence that gender diversity or skill composition affect market capitalization or short-term debt over the short time horizon was found. This suggests that any potential effect of board or workforce characteristics is either too small to detect in the short run or operates through longer-term strategic adjustments that cannot be captured within the limited time horizon of the dataset.

To explore whether heterogeneous effects of gender diversity and skill ratio emerge across structurally different types of firms or in response to aggregate conditions, several interaction models were estimated. The main regressors were interacted with baseline firm size (proxied by average employment), with the initial share of women on the board, with firm age, and with a 2007 dummy capturing the macroeconomic shift immediately before the financial crisis. Tables 7 and 8 report the corresponding results.

Overall, the evidence points to very limited heterogeneity. None of the interaction terms are statistically significant in either outcome equation, indicating that the estimated effects of gender diversity and skill composition do not differ systematically across these firm characteristics. An exception is detected for market capitalization: the interaction between the skill ratio and the 2007 dummy is positive and statistically significant. This result suggests that, in the year preceding the financial crisis, firms with a more skilled workforce enjoyed a modest valuation premium compared to the previous year, which is consistent with the idea that investors rewarded human capital-intensive firms in that period.

Taken together, the heterogeneous-effects analysis supports the main conclusion: the lack of significance in the original models is not masking group-specific patterns or conditional effects; rather, it reflects the broader absence of within-firm variation in board composition and skill structure over the sample period.

In addition, the robustness checks consistently confirm that the main results are not sensitive to alternative specifications. Adding time fixed effects did not change the coefficients of gender diversity and skill ratio, indicating that common macroeconomic shocks do not bias these estimates. Random-effects models produced similar patterns, and the Hausman test confirmed that fixed effects are required for short-term debt, while both estimators are valid for market capitalization.

This lack of significance, however, does not necessarily invalidate the original hypotheses that gender board composition and skill ratio increase market capitalization and reduce short-term debt, since the analysis is constrained by several limitations. Most critically, the dataset only covers two consecutive years, which severely restricts the within-firm variation needed for fixed-effects identification. As previously presented in figure 5 and figure 6, gender board diversity and skill ratio are highly persistent over this short horizon, leaving very limited changes from which to estimate potential effects.

A second limitation concerns potential endogeneity. Even though board composition is largely stable in the short run, it may still be influenced by longer-run performance trends, expectations or strategic adjustments: mechanisms that cannot be addressed with the available data and instruments. The absence of richer governance-related controls further restricts the ability to isolate the true channels through which board gender diversity or worker skill might operate.

Finally, the sample ends in 2007, right before the financial crisis. This means that crisis-related dynamics such as tightening credit conditions, shift in investors' sentiment and changes in risk-taking are not captured. During periods of stress, differences in managerial prudence and internal capabilities tend to have a broader impact in both financing decisions and market valuation. With richer variation in both

firm outcomes and macro-financial conditions, it would have been possible to test whether more board diverse and skilled firms were better at preserving their valuation and adjusting the capital structure in adverse financial conditions.

Taken together, these limitations suggest that the absence of significant effects in the regressions may reflect data constraints rather than the absence of any underlying relationship between board composition, workforce skill and firm outcomes. Longer or more dynamic datasets would be required to assess these mechanisms more reliably.

Conclusion

The results raise an important question: to what extent are firms' financing choices and valuations shaped by external forces rather than internal governance characteristics in the short-run? Given the limited year-to-year movement in gender diversity and skill ratio, the study suggests that a short-run variation of these internal characteristics may simply be too small to exert measurable influence on outcomes. External forces such as interest rate conditions, shifts in credit supply and macroprudential regulations could play a more determinant role in assessing access to short-term borrowing. Furthermore, macroeconomic uncertainty, sector-specific shocks and fiscal policies may exert a more rapid impact on market valuation rather than the gradual adjustment in board composition and workforce skills. These considerations are particularly important in light of the period covered by the dataset. The years 2006–2007 were characterized by abundant liquidity, loosened credit conditions and the buildup of financial imbalances preceding the global financial crisis. Such macroeconomic environment may have dominated firms' financing and valuation dynamics, reducing the relative importance of incremental governance changes and workforce skill composition, overshadowing their effect within the two-year panel. Future research would benefit from using longer time frames that allow governance characteristics and workforce composition to evolve meaningfully over time, as well as from incorporating external firm-level information, such as interest rates faced by firms, bank-firm relationships and credit constraints. Extending the analysis to post-crisis periods, when governance reforms became widespread, may also reveal effects that cannot be detected in the short pre-crisis window. Finally, investigating how internal and external factors interact, for example whether more diverse boards and more skilled workforce adjust differently to credit shocks, could provide deeper insights into the relationships.

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Appendix: graphs

Figure 1: Log market capitalization distribution

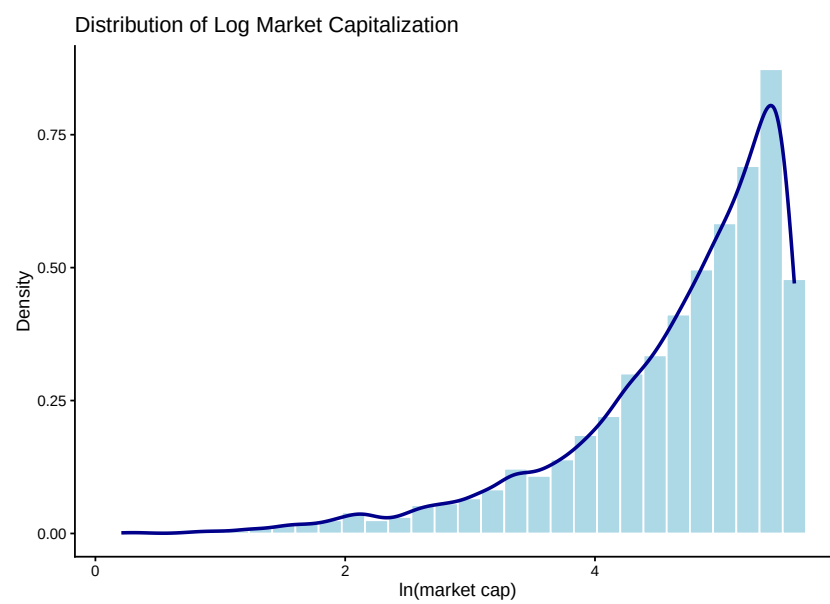


Figure 2: Centered short-term debt distribution

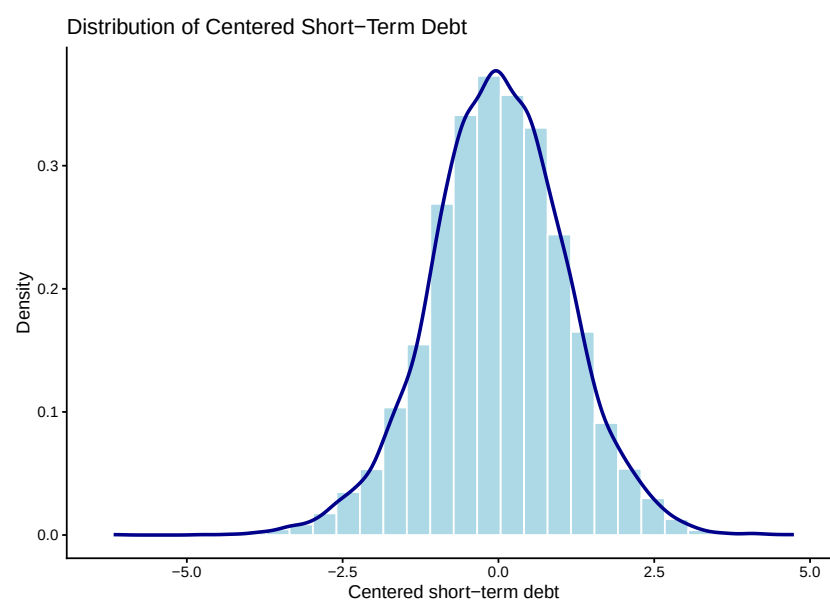


Figure 3: Gender diversity distribution

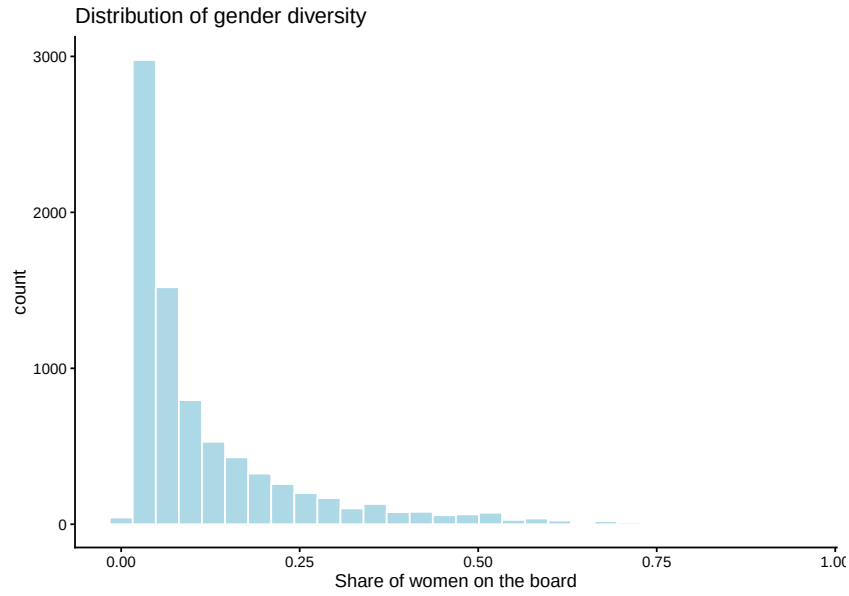
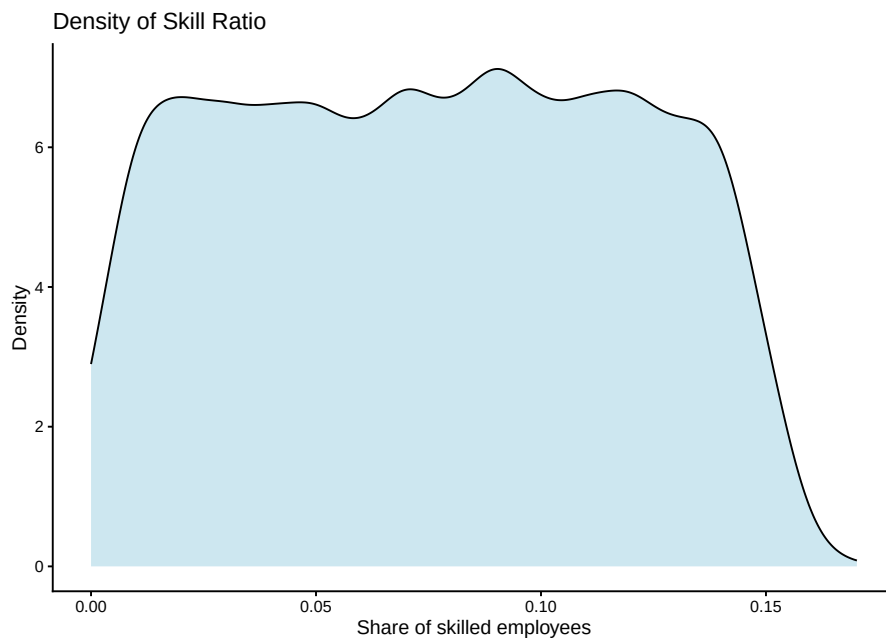


Figure 4: Density of the skill ratio



Notes: The skill ratio is a bounded variable defined as the share of skilled employees within each firm. Because the variable takes discrete repeated values and exhibits little within-firm variation over the two-year panel, a kernel density provides a clearer representation of its distribution than a histogram, which would largely reproduce identical spikes.

Figure 6: Within-firm skill ratio changes the distribution

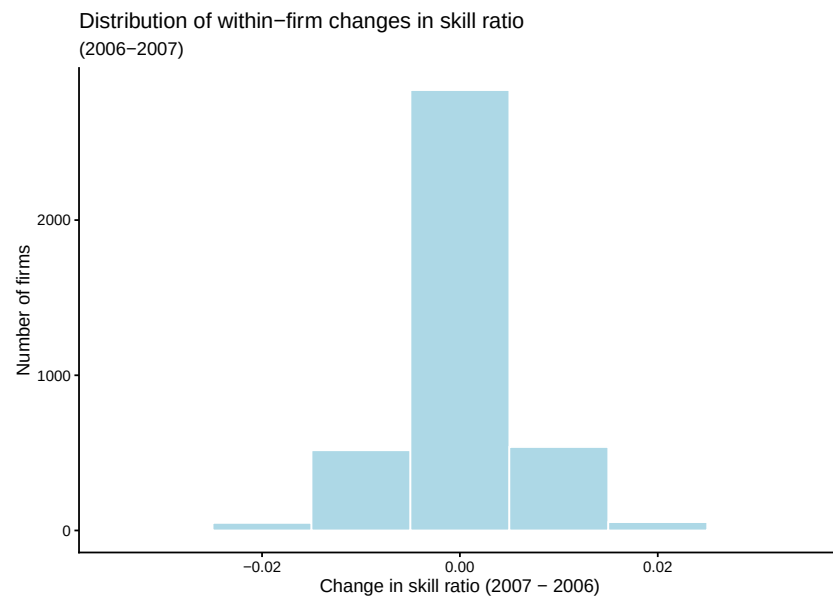


Figure 5: Within-firm gender diversity changes distribution

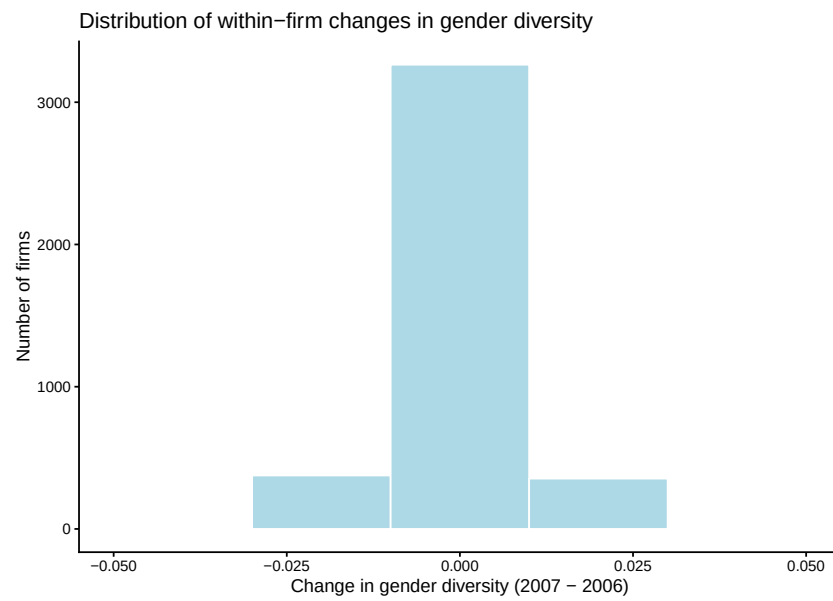


Figure 7: Boxplot log market capitalization distribution by sector

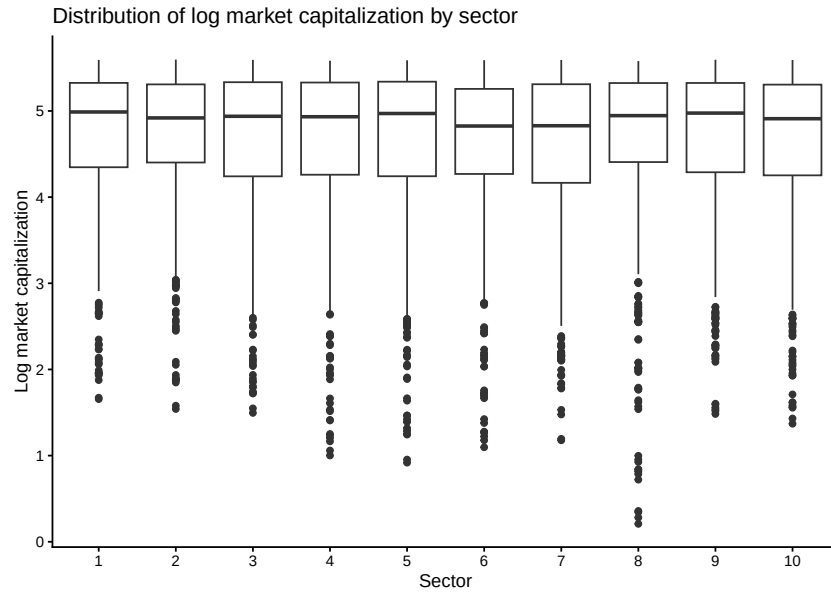
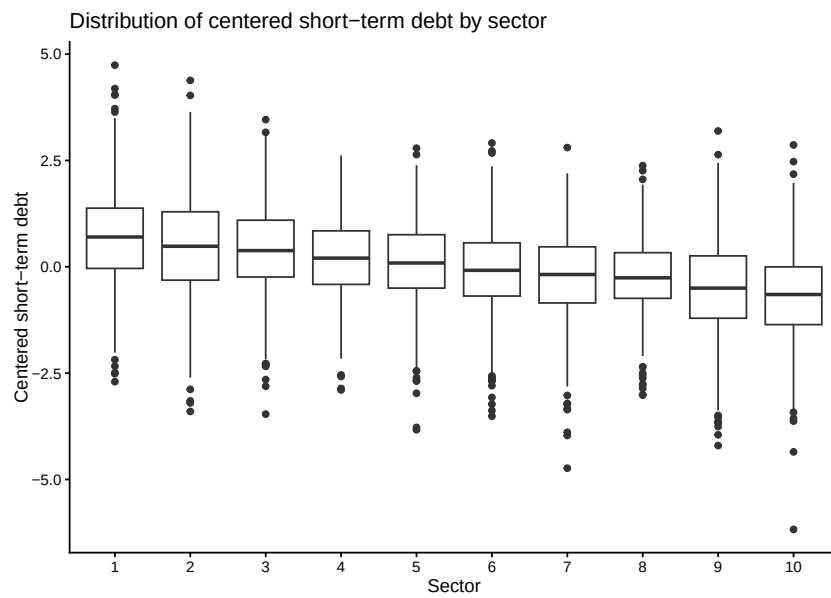


Figure 8: Boxplot centered short-term debt distribution by sector



Appendix: tables

Table 2: OLS regressions: simple and extended specifications

	Log market capitalization			Centered short-term debt		
	(1)	(2)	(3)	(4)	(5)	(6)
Gender diversity	0.026 (0.072)	—	0.082 (0.077)	−2.426*** (0.086)	—	−1.546*** (0.088)
Skill ratio	—	−0.462** (0.224)	−0.459** (0.225)	—	−0.088 (0.283)	0.048 (0.257)
Age	—	—	0.001 (0.001)	—	—	−0.00000 (0.001)
Constant	4.663*** (0.013)	4.701*** (0.020)	4.711 (0.050)	0.296 (0.016)	0.007 (0.025)	0.764 (0.057)
Sector controls	No	No	Yes	No	No	Yes
Observations	8,000	8,000	8,000	8,000	8,000	8,000
R ²	0.00002	0.00053	0.003	0.090	0.00001	0.176

Notes: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Models (1)–(3) estimate log market capitalization with: (1) gender diversity; (2) skill ratio; (3) both skill ratio and gender diversity with age and sector controls. Models (4)–(6) estimate centered short-term debt with the same structure.

Table 3: Summary statistics of the dataset

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
id	1	1001	2000	2000	3000	4000
sector	1	3	6	5.587	8	10
year	2006	2006	2006	2006	2007	2007
emp	104	9452	18249	18263	27262	35868
age	2	23	30	29.88	37	67
profit	−3.38	−0.55	0.43	0.94	1.93	17.97
inv	0	16.98	22.49	22.29	29.88	62
mktcap	1.231	72.736	137.876	137.697	204.155	269.451
shortdebt	15.82	21.30	21.99	22.00	22.71	26.74
nboard	82	97	100	99.98	103	118
wboard	0	4	7	12.19	16	98
skemp	0	388	1042	1371	2074	5719
genderdiversity	0.00000	0.03810	0.06667	0.12193	0.15686	0.93750
skillratio	0.00000	0.03999	0.07997	0.07572	0.11002	0.17021
lnmktcap	0.2077	4.2868	4.9264	4.6664	5.3189	5.5964
shortdebtcentrage	−6.174205	−0.696194	−0.002675	0.000000	0.712601	4.740234

Notes: The dataset is a balanced panel of 4,000 firms observed in 2006 and 2007 (8,000 firm-year observations). *genderdiversity* is defined as the share of women on the board, *skillratio* as the share of skilled employees, and *shortdebtcentrage* is short-term debt demeaned by its sample mean. The variable *lnmktcap* is the natural logarithm of market capitalization. All monetary variables are expressed in millions of euros. Sector ranges from 1 to 10.

Table 4: One-way firm fixed effects regression results

	<i>Dependent variable:</i>	
	Log market capitalization	Centered short-term debt
	(1)	(2)
Gender diversity	0.009 (0.016)	-0.869 (2.543)
Skill ratio	0.019 (0.020)	1.627 (3.689)
Age	0.0002 (0.0001)	1.021*** (0.019)
Observations	8,000	8,000
R ²	0.001	0.428
Adjusted R ²	-1.000	-0.144

Firm fixed effects included; no time fixed effects. Standard errors clustered at the firm level.

Centered short-term debt is expressed in millions and mean-centered at the firm level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Random-effects regression results

	<i>Dependent variable:</i>	
	Log market capitalization	Centered short-term debt
	(1)	(2)
Gender diversity	0.011 (0.016)	-1.546*** (0.097)
Skill ratio	0.017 (0.020)	0.050 (0.222)
Age	0.0002 (0.0001)	0.0002 (0.001)
Sector 2	-0.001 (0.058)	-0.192*** (0.052)
Sector 3	-0.050 (0.061)	-0.252*** (0.047)
Sector 4	-0.067 (0.061)	-0.421*** (0.044)
Sector 5	-0.065 (0.063)	-0.507*** (0.046)
Sector 6	-0.089 (0.060)	-0.652*** (0.046)
Sector 7	-0.121** (0.060)	-0.753*** (0.047)
Sector 8	-0.041 (0.062)	-0.777*** (0.042)
Sector 9	-0.026 (0.059)	-0.991*** (0.052)
Sector 10	-0.073 (0.061)	-1.140*** (0.049)
Constant	4.712 (0.042)	0.757 (0.051)
Observations	8,000	8,000
R ²	0.001	0.169
Adjusted R ²	-0.0003	0.168

Random-effects panel estimators; sector included as dummies, sector 1 being the baseline category
Standard errors clustered at the firm level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Hausman Tests: Fixed Effects vs. Random Effects

Model	Test Statistic (χ^2)	p-value
Log market capitalization	1.8716	0.5995
Centered short-term debt	3006.1	$< 2.2 \times 10^{-16}$

Notes: The Hausman test examines the null hypothesis that the Random Effects (RE) estimator is consistent and efficient, implying no systematic difference between the RE and FE coefficients. A significant p-value leads to rejection of the null, indicating that the RE estimator is inconsistent due to correlation between the regressors and the unobserved firm-specific effects. In this case, the FE specification is preferred as it provides consistent estimates.

Interaction equations for log market capitalization

$$\begin{aligned}
\ln(mktcap_{it}) &= \beta_1 \text{genderdiversity}_{it} + \beta_2 \text{skillratio}_{it} + \beta_3 \text{age}_{it} \\
&\quad + \beta_4(\text{genderdiversity}_{it} \times \text{base_emp}_i) + \beta_5(\text{skillratio}_{it} \times \text{base_emp}_i) + \mu_i + u_{it} \\
\ln(mktcap_{it}) &= \beta_1 \text{genderdiversity}_{it} + \beta_2 \text{skillratio}_{it} + \beta_3 \text{age}_{it} \\
&\quad + \beta_4(\text{genderdiversity}_{it} \times \text{base_gender}_i) + \mu_i + u_{it} \\
\ln(mktcap_{it}) &= \beta_1 \text{genderdiversity}_{it} + \beta_2 \text{skillratio}_{it} + \beta_3 \text{age}_{it} \\
&\quad + \beta_4(\text{genderdiversity}_{it} \times \text{age}_{it}) + \mu_i + u_{it} \\
\ln(mktcap_{it}) &= \beta_1 \text{genderdiversity}_{it} + \beta_2 \text{skillratio}_{it} + \beta_3 \text{age}_{it} + \beta_4 \text{year2007}_t \\
&\quad + \beta_5(\text{genderdiversity}_{it} \times \text{year2007}_t) + \beta_6(\text{skillratio}_{it} \times \text{year2007}_t) + \mu_i + u_{it}
\end{aligned}$$

Table 7: Fixed-effects interaction models for log market capitalization

	Size	Baseline gender	Age interaction	2007 shock
Gender diversity	0.0397 (0.0635)	0.0176 (0.0216)	-0.0111 (0.0259)	0.0068 (0.0164)
Skill ratio	0.0729 (0.0784)	0.0191 (0.0201)	0.0192 (0.0201)	0.0139 (0.0200)
Age	0.00018 (0.00013)	0.00018 (0.00013)	0.00010 (0.00016)	0.00018 (0.00013)
Gend.div $\times$ emp	-1.69×10^{-6} (2.61e-6)	—	—	—
Skill $\times$ emp	-2.92×10^{-6} (3.19e-6)	—	—	—
Gend.div $\times$ baseline gend.	—	-0.0534 (0.0906)	—	—
Gend.div $\times$ age	—	—	0.00069 (0.00079)	—
Year 2007	—	—	—	-0.00056* (0.00026)
Gend.div $\times$ 2007	—	—	—	0.00132 (0.00101)
Skill $\times$ 2007	—	—	—	0.00768** (0.00273)

Notes: Robust SE in parentheses. Size proxied by average employees.

Interaction equations correspond to the four specifications listed above.

* $p < 0.05$, ** $p < 0.01$.

Interaction equations for centered short-term debt

$$\begin{aligned} shortdebtcentrage_{it} = & \gamma_1 genderdiversity_{it} + \gamma_2 skillratio_{it} + \gamma_3 age_{it} \\ & + \gamma_4 (genderdiversity_{it} \times base_emp_i) + \gamma_5 (skillratio_{it} \times base_emp_i) + \mu_i + \varepsilon_{it} \end{aligned}$$

$$\begin{aligned} shortdebtcentrage_{it} = & \gamma_1 genderdiversity_{it} + \gamma_2 skillratio_{it} + \gamma_3 age_{it} \\ & + \gamma_4 (genderdiversity_{it} \times base_gender_i) + \mu_i + \varepsilon_{it} \end{aligned}$$

$$\begin{aligned} shortdebtcentrage_{it} = & \gamma_1 genderdiversity_{it} + \gamma_2 skillratio_{it} + \gamma_3 age_{it} \\ & + \gamma_4 (genderdiversity_{it} \times age_{it}) + \mu_i + \varepsilon_{it} \end{aligned}$$

$$\begin{aligned} shortdebtcentrage_{it} = & \gamma_1 genderdiversity_{it} + \gamma_2 skillratio_{it} + \gamma_3 age_{it} + \gamma_4 year2007_t \\ & + \gamma_5 (genderdiversity_{it} \times year2007_t) + \gamma_6 (skillratio_{it} \times year2007_t) + \mu_i + \varepsilon_{it} \end{aligned}$$

Table 8: Fixed-effects interaction models for centered short-term debt

	Size	Baseline gender	Age interaction	2007 shock
Gender diversity	2.4696 (5.1981)	-4.0097 (3.4914)	0.6255 (4.8947)	-0.6839 (2.5516)
Skill ratio	6.8424 (7.4379)	1.6134 (3.6854)	1.6182 (3.6919)	1.7512 (3.7023)
Age	1.0208*** (0.0186)	1.0201*** (0.0186)	1.0271*** (0.0231)	1.0562*** (0.0405)
Gend.div × emp	-0.00018 (0.00026)	—	—	—
Skill × emp	-0.00028 (0.00035)	—	—	—
Gend.div × baseline gend.	—	18.7888 (18.5848)	—	—
Gend.div × age	—	—	-0.0519 (0.1448)	—
Gend.div × 2007	—	—	—	-0.1501 (0.1628)
Skill × 2007	—	—	—	-0.2261 (0.4313)

Notes: Robust SE in parentheses. Size proxied by average employees.

Interaction equations correspond to the four specifications listed above.

*** $p < 0.01$.

bottomrule

Appendix: R Code

```
1 # loading libraries
2 library(plm)
3 library(sandwich)
4 library(lmtest)
5 library(stargazer)
6 library(dplyr)
7 library(tidyr)
8 library(ggplot2)
9 library(patchwork)
10
11 # SIMPLE OLS
12 # OLS market cap - gender diversity
13 lm1 <- lm(lnmktcap ~ genderdiversity, data = pgfc2008)
14 summary(lm1)
15 # OLS short-term debt and gender diversity
16 lm2 <- lm(shortdebtcentrage ~ genderdiversity, data = pgfc2008)
17 summary(lm2)
18 # OLS market cap - skill ratio
19 lm3 <- lm(lnmktcap ~ skillratio, data = pgfc2008)
20 summary(lm3)
21 # OLS short-term debt - skill ratio
22 lm4 <- lm(shortdebtcentrage ~ skillratio, data = pgfc2008)
23 summary(lm4)
24
25 # EXTENDED OLS
26 # sector is turned as factor variable
27 pgfc2008$sector <- factor(pgfc2008$sector)
28 # regressions
29 lm5 <- lm(lnmktcap ~ genderdiversity + age + sector, data = pgfc2008)
30 summary(lm5)
31 lm6 <- lm(shortdebtcentrage ~ genderdiversity + age + sector, data = pgfc2008)
32 summary(lm6)
33 lm7 <- lm(lnmktcap ~ skillratio + age + sector, data = pgfc2008)
34 summary(lm7)
35 lm8 <- lm(shortdebtcentrage ~ skillratio + age + sector, data = pgfc2008)
36 summary(lm8)
37 # joint variables of interest
38 lm9 <- lm(lnmktcap ~ genderdiversity + skillratio + age + sector, data = pgfc2008)
39 summary(lm9)
40 lm10 <- lm(shortdebtcentrage ~ genderdiversity + skillratio + age + sector, data =
41             pgfc2008)
42 summary(lm10)
43
44 # dataset summary statistics
45 pgfc2008_clean <- subset(pgfc2008, select = -c(high_genderdiv, high_shortdebt, year_bin,
46             dummy1, dummy2))
47 summary(pgfc2008_clean)
48 # distribution checks
49 # log market cap
50 # Histogram + density overlay
51 ggplot(pgfc2008, aes(x = lnmktcap)) +
52   geom_histogram(aes(y = ..density..),
53     bins = 30, fill = "lightblue", color = "white") +
54   geom_density(color = "darkblue", lwd = 1) +
```

```

53   labs(title = "Distribution of Log Market Capitalization",
54         x = "ln(market cap)", y = "Density") +
55   theme_classic()
56 # centered-short term debt
57 # Histogram + density overlay
58 ggplot(pgfc2008, aes(x = shortdebtcentrage)) +
59   geom_histogram(aes(y = ..density..),
60                 bins = 30, fill = "lightblue", color = "white") +
61   geom_density(color = "darkblue", lwd = 1) +
62   labs(title = "Distribution of Centered Short-Term Debt",
63         x = "Centered short-term debt", y = "Density") +
64   theme_classic()
65 # gender diversity distribution
66 ggplot(pgfc2008, aes(x = genderdiversity)) +
67   geom_histogram(fill = "lightblue", color = "white") +
68   labs(title = "Distribution of gender diversity",
69         x = "Share of women on the board") +
70   theme_classic()
71
72 # DENSITY plot skill ratio
73 ggplot(pgfc2008, aes(x = skillratio)) +
74   geom_density(fill="lightblue", alpha=0.6) +
75   labs(title="Density of Skill Ratio",
76         x="Share of skilled employees",
77         y="Density") +
78   theme_classic()
79
80 # BOXPLOT market capitalization by sector
81 ggplot(pgfc2008, aes(x = factor(sector), y = lnmktcap)) +
82   geom_boxplot() +
83   labs(
84     title = "Distribution of log market capitalization by sector",
85     x = "Sector",
86     y = "Log market capitalization"
87   ) +
88   theme_classic()
89 # BOXPLOT centered short-term debt by sector
90 ggplot(pgfc2008, aes(x = factor(sector), y = shortdebtcentrage)) +
91   geom_boxplot() +
92   labs(
93     title = "Distribution of centered short-term debt by sector",
94     x = "Sector",
95     y = "Centered short-term debt"
96   ) +
97   theme_classic()
98
99 # ONE WAY FIXED EFFECT MODEL
100 # log market cap + controls
101 fe_mkt <- plm(lnmktcap ~ genderdiversity + skillratio + age,
102              data = pgfc2008, index = c("id", "year"), model = "within")
103 # short term debt centrage + controls
104 fe_std <- plm(shortdebtcentrage ~ genderdiversity + skillratio + age,
105              data = pgfc2008, index = c("id", "year"), model = "within")
106 # adding cluster-robust SE (at firm level)
107 robust_mkt <- vcovHC(fe_mkt, type = "HC1", cluster = "group")
108 robust_std <- vcovHC(fe_std, type = "HC1", cluster = "group")

```

```

109 se_mkt <- sqrt(diag(robust_mkt))
110 se_std <- sqrt(diag(robust_std))
111 # coefficient table
112 coeftest(fe_mkt, vcov = robust_mkt)
113 coeftest(fe_std, vcov = robust_std)
114
115 # TWO WAYS FIXED EFFECT MODEL
116 # Log market cap with firm + year FE
117 fe_mkt_2w <- plm(lnmktcap ~ genderdiversity + skillratio + factor(year),
118                 data = pgfc2008, index = c("id", "year"), model = "within")
119 # Short-term debt with firm + year FE
120 fe_std_2w <- plm(shortdebtcentrage ~ genderdiversity + skillratio + factor(year), data
121                 = pgfc2008, index = c("id", "year"),
122                 model = "within")
123 # Robust SE (clustered at firm level)
124 robust_mkt_2w <- vcovHC(fe_mkt_2w, type = "HC1", cluster = "group")
125 robust_std_2w <- vcovHC(fe_std_2w, type = "HC1", cluster = "group")
126 se_mkt_2w <- sqrt(diag(robust_mkt_2w))
127 se_std_2w <- sqrt(diag(robust_std_2w))
128 # Coefficient table:
129 coeftest(fe_mkt_2w, vcov = robust_mkt_2w)
130 coeftest(fe_std_2w, vcov = robust_std_2w)
131
132 # DISTRIBUTION OF WITHIN-FIRM CHANGES: skill ratio
133 # changes in skill ratio from 2006 o 2007
134 skill_wide <- pgfc2008 %>% # Make wide format
135   select(id, year, skillratio) %>%
136   filter(year %in% c(2006, 2007)) %>%
137   pivot_wider(
138     names_from = year,
139     values_from = skillratio,
140     names_prefix = "skill_"
141   ) %>%
142   filter(!is.na(skill_2006), !is.na(skill_2007))
143 # Compute the change
144 skill_diff <- skill_wide %>%
145   mutate(delta_skill = skill_2007 - skill_2006)
146 # Plot histogram of the change
147 ggplot(skill_diff, aes(x = delta_skill)) +
148   geom_histogram(binwidth = 0.01, fill="lightblue", color="white") +
149   labs(
150     title = "Distribution of within-firm changes in skill ratio",
151     subtitle = "(2006 2007 )",
152     x = "Change in skill ratio (2007 - 2006)",
153     y = "Number of firms"
154   ) +
155   theme_classic()
156
157 # DISTRIBUTION OF WITHIN-FIRM CHANGES: genderdiversity
158 # changes in gender diversity from 2006 to 2007
159 gender_wide <- pgfc2008 %>%
160   select(id, year, genderdiversity) %>%
161   filter(year %in% c(2006, 2007)) %>%
162   pivot_wider(
163     names_from = year,
164     values_from = genderdiversity,

```

```

164     names_prefix = "gender_"
165   )
166   gender_diff <- gender_wide %>%
167     mutate(delta_gender = gender_2007 - gender_2006)
168   ggplot(gender_diff, aes(x = delta_gender)) +
169     geom_histogram(binwidth = 0.02, color = "white", fill = "lightblue") +
170     labs(
171       title = "Distribution of within-firm changes in gender diversity",
172       x = "Change in gender diversity (2007 - 2006)",
173       y = "Number of firms"
174     ) +
175     theme_classic()
176
177
178 # RANDOM FIXED EFFECTS
179 class(pgfc2008$sector) # making sure sector is a factor
180 # log market cap - RE
181 re_mkt <- plm(lnmktcap ~ genderdiversity + skillratio + age + sector,
182               data = pgfc2008, index = c("id", "year"), model = "random")
183 # short term debt - RE
184 re_std <- plm(shortdebtcentrage ~ genderdiversity + skillratio + age + sector,
185               data = pgfc2008, index = c("id", "year"), model = "random")
186 # cluster robust standard errors
187 robust_re_mkt <- vcovHC(re_mkt, type = "HC1", cluster = "group")
188 robust_re_std <- vcovHC(re_std, type = "HC1", cluster = "group")
189 se_re_mkt <- sqrt(diag(robust_re_mkt))
190 se_re_std <- sqrt(diag(robust_re_std))
191 # coefficient table
192 coeftest(re_mkt, vcov = robust_re_mkt)
193 coeftest(re_std, vcov = robust_re_std)
194
195 # Hausman tests: FE vs RE
196 hausman_mkt <- phtest(fe_mkt, re_mkt)
197 hausman_std <- phtest(fe_std, re_std)
198 hausman_mkt
199 hausman_std
200
201 # TESTING HETEROGENOUS EFFECTS through interaction terms
202 pgfc2008$base_emp <- ave(pgfc2008$emp, pgfc2008$id,
203                           FUN = function(x) mean(x, na.rm = TRUE)) #baseline firm size (
204                               proxied by employees)
205 pgfc2008$base_gender <- ave(pgfc2008$genderdiversity, pgfc2008$id,
206                               FUN = function(x) mean(x, na.rm = TRUE)) #baseline gender
207                               diversity
208 pgfc2008$year2007 <- as.numeric(pgfc2008$year == 2007) # 2007 year dummy
209
210 # LOG MARKET CAPITALIZATION
211 # testing if the effect of gender diversity and skill ratio varies with firm size:
212 fe_inter_size_mkt <- plm(
213   lnmktcap ~ genderdiversity*base_emp + skillratio*base_emp + age,
214   data = pgfc2008,
215   index = c("id", "year"),
216   model = "within")
217 coeftest(fe_inter_size_mkt,
218           vcov = vcovHC(fe_inter_size_mkt, type = "HC1", cluster = "group"))
219 # testing if the effect differs between firms with already more women on board:

```

```

218 fe_inter_gdiv_mkt <- plm(
219   lnmktcap ~ genderdiversity*base_gender + skillratio + age,
220   data = pgfc2008,
221   index = c("id","year"),
222   model = "within")
223 coeftest(fe_inter_gdiv_mkt,
224   vcov = vcovHC(fe_inter_gdiv_mkt, type = "HC1", cluster = "group"))
225 # heterogeneity by age:
226 fe_inter_age_mkt <- plm(
227   lnmktcap ~ genderdiversity*age + skillratio + age,
228   data = pgfc2008,
229   index = c("id","year"),
230   model = "within")
231 coeftest(fe_inter_age_mkt,
232   vcov = vcovHC(fe_inter_age_mkt, type = "HC1", cluster = "group"))
233 # testing if the effect changed with the macroeconomic shock: interaction between
234 # skillratio and genderdiversity and the 2007 dummy.
235 fe_inter_year_mkt <- plm(
236   lnmktcap ~ genderdiversity*year2007 + skillratio*year2007 + age,
237   data = pgfc2008,
238   index = c("id","year"),
239   model = "within")
240 coeftest(fe_inter_year_mkt,
241   vcov = vcovHC(fe_inter_year_mkt, type = "HC1", cluster = "group"))
242
243 # SHORT-TERM DEBT
244 # testing if the effect of gender diversity and skill ratio varies with firm size:
245 fe_inter_size_debt <- plm(
246   shortdebtcentrage ~ genderdiversity*base_emp + skillratio*base_emp + age,
247   data = pgfc2008,
248   index = c("id","year"),
249   model = "within")
250 coeftest(fe_inter_size_debt,
251   vcov = vcovHC(fe_inter_size_debt, type = "HC1", cluster = "group"))
252 # testing if the effect differs between firms with already more women on board:
253 fe_inter_gdiv_debt <- plm(
254   shortdebtcentrage ~ genderdiversity*base_gender + skillratio + age,
255   data = pgfc2008,
256   index = c("id","year"),
257   model = "within")
258 coeftest(fe_inter_gdiv_debt,
259   vcov = vcovHC(fe_inter_gdiv_debt, type = "HC1", cluster = "group"))
260 # heterogeneity by age:
261 fe_inter_age_debt <- plm(
262   shortdebtcentrage ~ genderdiversity*age + skillratio + age,
263   data = pgfc2008,
264   index = c("id","year"),
265   model = "within")
266 coeftest(fe_inter_age_debt,
267   vcov = vcovHC(fe_inter_age_debt, type = "HC1", cluster = "group"))
268 # testing if the effect changed with the macroeconomic shock: interaction between
269 # skillratio and genderdiversity and the 2007 dummy.
270 fe_inter_year_debt <- plm(
271   shortdebtcentrage ~ genderdiversity*year2007 + skillratio*year2007 + age,
272   data = pgfc2008,
273   index = c("id","year"),

```

```
274     model = "within")
275     coeftest(fe_inter_year_debt,
276             vcov = vcovHC(fe_inter_year_debt, type = "HC1", cluster = "group"))
```